

Padala SriHarsha

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Professional Summary

Recent Artificial Intelligence and Data Science graduate with hands-on experience in Machine Learning, Deep Learning, NLP, Generative AI, and Explainable AI. Skilled in Python, PyTorch, FastAPI, Scikit-learn, and Transformer-based architectures. Experienced in developing recommendation systems, image captioning models, and predictive analytics solutions. Strong understanding of Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Databases, and AI Agent concepts. Passionate about building intelligent AI systems that solve real-world business problems.

Education

Chaitanya Bharathi Institute of Technology (CBIT)

2022 – 2025

B.E. in Artificial Intelligence & Data Science

CGPA: 7.7/10

TKR College of Engineering

2019 – 2022

Diploma in Engineering

GPA: 8.7/10

Technical Skills

Programming Languages: Python, SQL, R, Java

Machine Learning: Regression, Classification, Clustering, Ensemble Models, Feature Engineering

Deep Learning: CNNs, Transformers, Sequence Models, Transfer Learning

Generative AI: Large Language Models (LLMs), Prompt Engineering, RAG Concepts, AI Agents

NLP: Tokenization, Embeddings, Text Classification, NER, Semantic Search, Information Extraction

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Hugging Face Transformers

AI Frameworks: LangChain (Basics), LlamaIndex (Basics)

MLOps & Deployment: Docker, Git, GitHub Actions, FastAPI, MLflow (Basic), CI/CD Concepts

Databases: MySQL, SQL, ChromaDB (Basic), Vector Database Concepts

Cloud Platforms: Google Cloud Platform (Basic), AWS Fundamentals

Tools: Power BI, Excel, Jupyter Notebook, Google Colab

Projects

Deep Learning for Automated Image Captioning

PyTorch — CNN — Transformers — NLP

- Developed an image captioning system using ResNet encoder and Transformer decoder architecture.
- Implemented tokenization, vocabulary creation, sequence padding, and custom dataloaders.
- Achieved BLEU-4 score of 0.62 on Flickr8k dataset.
- Published research work in IEEE IDCIoT 2025 conference.

Online Retail Recommendation System

Python — Collaborative Filtering — ML

- Built a recommendation engine using customer purchase history data.
- Implemented collaborative filtering and matrix factorization techniques.
- Improved recommendation quality through hyperparameter tuning and cross-validation.
- Generated personalized product recommendations across 50K+ transactions.

- Performed EDA and preprocessing on meteorological datasets.
- Built Random Forest model achieving 94% prediction accuracy.
- Conducted feature importance analysis and model validation.
- Visualized seasonal and spatial fire-risk trends.

Internship Experience

DRDL, Ministry of Defence

Internship

- Worked on AI and data-driven applications involving machine learning workflows.
- Assisted in data preprocessing, model development, testing, and performance evaluation.
- Collaborated with technical teams to analyze and solve real-world engineering problems.

Publications

- IEEE Conference Publication: *Deep Learning for Automated Image Captioning*, IDCIoT 2025, REVA University.

Certifications

- Data Science Certification – Infosys Springboard
- Machine Learning Fundamentals
- Deep Learning and NLP Courses
- NCC 'A' Certificate